Zixuan Li

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EDUCATION

The Hong Kong University of Science and Technology 08/2026 – 06/2027 (Incoming)
M.Sc. in Big Data Technology Hong Kong, China

Jilin University 09/2022 – 07/2026
B.Eng. in Computer Science and Technology Changchun, China

- **GPA:** 88.79/100 (Top 13%)
- **Core Coursework:** Computer Architecture (96), Database & Security (A+), Algorithm Design (93.7), Compiler Principles (93.2), Artificial Intelligence, Operating Systems.

PUBLICATIONS

[1] **FLINT: Influence-Guided Active Learning Framework for LoRA via Curvature-Aware Data Selection and Fine-Tuning.** *Z. Li, S. Guo, et al.*; Accepted to the Conference on Language Modeling (COLM) 2026.

[2] **Multimodal Fake News Detection using Graph Neural Networks and Attention Mechanisms.** *Z. Li*; *Advances in Engineering Innovation*, 2024

RESEARCH EXPERIENCE

FLINT — Influence-Guided Active Learning for LoRA Fine-Tuning 12/2025 – 04/2026
Jilin University / Research Project | ***First Author***

- Designed **FLINT**, a two-stage active learning pipeline integrating entropy pre-filtering, **SVD** cold-start stabilization, and **K-FAC** curvature-aware influence scoring optimized for the LoRA manifold.
- Streamlined candidate evaluations by **70–80%** via adaptive entropy thresholding, successfully constraining the data selection to a strict **20%** budget across three active learning iterations.
- Achieved LLaMA-3.2-3B accuracies of **71.13% / 66.8% / 73.0%** on GSM8K / BBH / StrategyQA, consistently outperforming full-data LoRA baselines (**68.85% / 64.14% / 69.0%**).
- Accelerated end-to-end selection and training to **1.67h** on a single V100 GPU, yielding a **5.73**× speedup over LESS and **2.80**× over DataInf with a peak VRAM of **16.6GB**.
- **Accepted to **COLM 2026**** as first author.

Dimension-Mapped Near-Memory Computing for Sparse Attention Acceleration 05/2025 – 12/2025
Jilin University HPC Center | ***First Author (Advisor: Prof. Xiaohui Wei)***

- Proposed a novel **dimension-based dataflow mapping** (as opposed to standard token-level partitioning) for Near-Memory Processing (NMP) systems, enforcing strict load balancing across parallel DRAM bank arrays via hash-routed feature decoupling.
- Developed a trace-driven hardware-software co-simulation engine using `PyTorch hooks` and `SimPy` discrete-event modeling to simulate physical DRAM row-buffer conflicts under highly skewed, long-tail sparse attention masks.
- Achieved a **27.7%** reduction in end-to-end LLM inference step latency (6920 ns → 5000 ns) and minimized workload variance (**coefficient of variation dropped from 23.89% → 7.09%**).

Multimodal Fake News Detection via GNN and Cross Attention Mechanisms 12/2024 – 02/2025
Jilin University | ***First Author***

- Formulated a multimodal fusion network architecture combining `BERT` (textual), `CNN+CLIP` (visual), and `GraphSAGE` (social graph structure) with an adaptive cross-modal attention routing mechanism.
- Demonstrated robust performance gains over baseline models on `FakeNewsNet` and `Sina Weibo` benchmarks.
- Independently conceptualized the methodology, performed extensive ablation studies, and drafted the full manuscript published in *Advances in Engineering Innovation*.

INDUSTRY EXPERIENCE

Tencent — WeChat Headquarters (WeChat Agent Research) 05/2026 – Present
Research Intern

- **Algorithmic Optimization for Multi-Turn Tool Integration:** Investigated critical inference-latency bottlenecks in multi-turn, tool-augmented Large Language Models (LLMs); formulated an in-context tool composition mechanism within hierarchical skill abstractions, accelerating end-to-end response velocity by **2.0**× (from **20 s to 10 s**).
- **Topological Design of Agentic Execution Systems:** Conceptualized a novel hybrid architecture integrating heuristic routing, stateful workflows, and autonomous agentic fallbacks; conducted rigorous empirical analysis on its structural generalization and edge-case stability, yielding a state-of-the-art first-token latency of **5–6 s** under high-concurrency settings.

Ping An Technology — Semantics Laboratory (NLP Research Group) 12/2025 – 05/2026
Research Intern

- **Mitigating Catastrophic Forgetting in Knowledge Alignment:** Developed an interactive conversational decision framework; fine-tuned parameterized LLMs (`Qwen3-8B`) via synthetic, chain-of-thought alignment topologies to systematically evaluate and mitigate *catastrophic forgetting* in resource-constrained models.
- **Scalable High-Dimensional Representation Pipelines:** Engineered a robust dual-stage cascade infrastructure leveraging dense neural vector retrieval coupled with fine-grained LLM classification; expanded the evaluation space from **31 to 340 semantic classes** while maintaining an empirical **Precision/Recall** > **95%** across extreme long-tail class distributions.

SKILLS

Programming	Python, C/C++, Java, CUDA C, Verilog, SQL, Lua
Machine Learning	PyTorch, Hugging Face Ecosystem (Transformers, PEFT), Ray, DeepSpeed
LLM Infrastructure	ReAct Framework, Tool-Use, MCP, Dify, SSE/AG-UI Streaming, Distributed Training
Data Systems	Spark SQL, WeData, MySQL, Discrete-Event Simulation (`SimPy`)
Language	TOEFL iBT **90** (R27 / W24 / L21 / S19); retake planned for August 2026 (targeting **105+**)